Martin Moreno Jara

Computer Science and Engineering Student | Bogota, Colombia
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PROFESSIONAL SUMMARY

Computer Science and Engineering student who enjoys connecting systems: APIs, IoT devices and machine learning models. Hands-on with Python and deep learning (TensorFlow/Keras), and used to working with AI assistants daily through a spec-driven development (SDD) workflow. Seeking a 2027 internship building AI solutions and integrations.

SKILLS

Programming Languages: Python, JavaScript/TypeScript, Java, SQL
AI and Data: TensorFlow, Keras, PyTorch, scikit-learn, NumPy, Pandas, Google Gemini
Frameworks and Libraries: FastAPI, Flask, Express.js, Node.js, React
Tools and Infrastructure: Git, GitHub, Linux, Docker, Firebase, Google Cloud Platform, Excel, Power BI
Databases: MongoDB, SQLite
Concepts: Machine Learning, REST APIs, Microservices, MQTT, Pub/Sub, TLS/SSL, Access Control, Spec-Driven Development, Scrum
Spoken Languages: Spanish (native), English (advanced), German (intermediate)

EDUCATION

Universidad Nacional de Colombia | Bogota, Colombia
B.Sc. in Computer Science and Engineering (Ingeniería de Sistemas y Computación) | Feb 2022 - Jun 2027 (Expected)

Relevant coursework: Intelligent Systems, Software Engineering and Architecture, Databases, Computer Networks.

PROJECTS

Brain Tumor MRI Classification | notebook | Mar 2025 - Jun 2025
Python, TensorFlow, Keras, Google Gemini

- In a 3-person team, compared three approaches to classify 5,700+ brain MRI scans into four classes: a custom CNN, EfficientNet transfer learning, and a zero-shot Google Gemini LLM baseline.
- Ran the EfficientNet transfer-learning experiments in TensorFlow/Keras, tuning hyperparameters across model sizes; the selected EfficientNetB3 reached a 98.9% combined accuracy/recall score on validation, against 72% for the Gemini baseline.
- Built the data pipeline: exploratory analysis, resizing, normalization, stratified train/validation split, and augmentation.

Intelligent Pot | firmware | dashboard | May 2026 - Jul 2026
C++ (Arduino), ESP32-S3, MQTT, HiveMQ

- Designed an IoT planter built on an ESP32-S3 that samples five environmental sensors (soil moisture, temperature, humidity, ambient light, air quality) and publishes telemetry over TLS-secured MQTT to a HiveMQ Cloud broker.
- Defined the MQTT topic contract between device and dashboard, using a retained state topic so a client connecting late receives the device's current configuration immediately.
- Automated irrigation and remote control over pub/sub: the dashboard sends watering, threshold, lighting and audio commands.
- Owned system design and integration across embedded, messaging, and web layers; wrote the docs and user manual.

Runpath | auth service | api gateway | Sep 2025 - Dec 2025
Python, FastAPI, Express.js, Nginx, Firebase, Docker, Google Cloud Platform

- Built an Express.js API Gateway and an Nginx reverse proxy that centralize request routing across microservices.
- Implemented an authentication backend with FastAPI and Firebase, handling identity management and access control.
- Configured encrypted communication between internal services using self-signed certificates to protect data in transit.
- Helped containerize the services with Docker and deploy them to Google Cloud Platform.